including damaged marks and IDs on challenging substrates. As semiconductor manufacturers expand production for AI infrastructure, advanced computing, and electronics, missed identification reads can delay processes, require manual intervention, and affect equipment utilisation. The In-Sight 1750 aims to address these issues by improving identification reliability and supporting traceability during semiconductor fabrication and packaging.
Cognex Corporation
http://cognex.com/en

A plug-in AI accelerator for existing processors

Ambarella has introduced the X7 edge AI accelerator, a chip designed to add AI processing to existing systems built around Arm and x86 host processors. It allows cameras, gateways, and industrial controllers to run additional AI workloads without redesigning the platform. The X7 targets OEMs and system integrators who want to add AI features to deployed equipment or evaluate new capabilities before developing next-generation hardware. It supports applications that combine computer vision, vision-language models, and natural-language interfaces. In current customer designs, the accelerator operates at two to five watts. It supports convolutional neural networks (CNNs), vision transformers, multimodal transformers, and hybrid networks using the company's third-generation CVflow computer vision and deep learning processor.



Ambarella
https://www.ambarella.com

Natural-language AI streamlines FPGA design

Lattice Semiconductor has introduced Lattice Prompt, an AI-based development tool for small and mid-range field-programmable gate arrays (FPGAs). It allows developers to use natural language for tasks across the FPGA design flow, including simulation, synthesis, mapping, placement and routing, timing analysis, and bitstream generation. The tool draws on Lattice design tools, documentation, datasheets, and its knowledge base to generate responses and support design decisions. It connects to large language models (LLMs) and integrated development environments (IDEs) through the Model Context Protocol (MCP). Lattice Prompt runs Lattice Radiant design software in the background and generates design reports and documentation. According to the company, it can improve productivity by 10 times or more on common design tasks. It is available as a free download.



Lattice Semiconductor
https://latticesemi.gcs-web.com

An indoor 3D LiDAR to catch intruders

Blickfeld has expanded its security product portfolio with the QbProtect S, a 3D LiDAR sensor for indoor security applications. It complements the QbProtect sensor, which is used for perimeter protection. The QbProtect S generates 3D point clouds with up to 1200 scan lines to monitor security zones with centimetre-level accuracy. It detects objects based on their size and direction of movement at distances of up to 35 metres. It can also detect reach-through attempts, such as a hand entering a protected area, at distances of up to 25 metres. The sensor processes point cloud data on-device in real-time.



Blickfeld
https://www.blickfeld.com

Motion sensors that stay steady through shocks and vibrations

STMicroelectronics has introduced two 6-axis inertial measurement units (IMUs), the LSM6DSK320X for consumer devices and the ISM6HGK256X for industrial applications. The sensors combine a gyroscope with two accelerometers to capture motion events across different acceleration ranges. The IMUs use a super-symmetric gyroscope architecture to reduce the effects of external vibrations and shocks. Combined with a differential signal chain and accelerometer tracking across a wide dynamic range, this architecture supports accurate event reconstruction. Target applications include fitness trackers, smart tags, industrial asset monitors, robots, and drones for photography and agricultural surveying, as well as accident alerts, tool-usage monitoring, robot teaching, sports-performance analysis, concussion detection, drone-flight stabilisation, structural-condition monitoring, and game controllers.



STMicroelectronics
https://www.st.com/content/st_com

A universal remote for home control and gaming

Hat & Hammer's OpenMote is a programmable, Arduino-compatible remote controller designed for smart home devices, consumer electronics, and gaming systems. The device uses an ESP32-S3 module with Wi-Fi, Bluetooth Low Energy (BLE), motion sensing, infrared communication, and audio capabilities. Of its 12 buttons, 11 are programmable, while a six-axis IMU detects motion. An infrared emitter and receiver, operating at 940 nm and 38 kHz respectively, allow it to learn and transmit remote-control signals for devices such as televisions and air conditioners. Developers can program OpenMote using the Arduino IDE, PlatformIO, or ESP-IDF.



Hat & Hammer (on Crowd Supply)
https://www.crowdsupply.com/hat-and-hammer